No Easy Wins: A Contamination-Controlled Benchmark for Evaluating Anomaly Detection and Model Selection

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Abstract

Generic anomaly detection methods are commonly evaluated on benchmarks that combine datasets containing normal observations and labeled anomalies. Designing such benchmarks is itself challenging. Existing datasets may contain duplicated anomalies, train/test leakage, anomalies that are trivially separable, or anomalies that are effectively indistinguishable from normal observations. Some also provide insufficient data for reliable comparisons, while commonly used evaluation metrics can be distorted by differences in anomaly prevalence. Reliable benchmarking therefore requires curated datasets, prevalence-aware evaluation, and statistically grounded comparisons between methods.

We introduce ADReal, a benchmark constructed by systematically curating and filtering datasets from multiple existing anomaly detection benchmarks. We remove duplicates, contaminated anomalies, leakage, and datasets that are either trivially easy or effectively impossible, yielding 151 datasets spanning tabular and multivariate time-series data. On these datasets, we evaluate roughly thirty widely used detectors, including classical isolation-, density-, distribution-, and projection-based methods and deep one-class and reconstruction-based models. Detectors are evaluated using a prevalence-aware metric that enables more comparable performance estimates across datasets with different anomaly rates.

ADReal also addresses a second problem: model selection. In real applications, anomalies are rare and heterogeneous, so a representative labeled anomaly sample may not be available for model selection or hyperparameter tuning. The detector must therefore often be selected using normal data alone. ADReal provides a controlled testbed for evaluating such normal-only selection methods: candidate detectors are selected without access to true anomalies, while their eventual performance on those anomalies provides an independent measure of selection quality. Using this testbed we evaluate popular unsupervised model-selection heuristics and release their results alongside the benchmark and detector evaluations.

Finally, we introduce SPARC, a normal-only model-selection method that generates synthetic anomalies from the normal data, by resampling a subset of each point's features from their own observed values so the combination is unusual while every value stays realistic, and ranks candidate detectors by how well they separate them. We benchmark every selection method against a strong reference the literature omits: always deploying the single detector with the best average performance. On ADReal this fixed detector is hard to beat, and the popular internal criteria, entropy and mass-volume, are significantly worse than it. SPARC is the only label-free method that significantly beats it, at regret 0.113 against 0.130 for the best fixed detector ($p = 0.04$), while beating entropy and mass-volume decisively (0.163 and 0.165 on the matched classical pool, $p < 0.001$), using a single probe and no tuning. We release ADReal, the complete detector evaluations, SPARC, and the associated code.

1. Introduction

New anomaly detection methods are introduced and compared on benchmarks that pair normal observations with labeled anomalies, drawn from fraud, industrial monitoring, health, and cybersecurity. The ranking such a benchmark produces is only as trustworthy as the data behind it. And because no single detector dominates (LOF wins on some datasets, KNN on others, HBOS on yet others), a practitioner must still choose one detector per dataset, usually with no labeled anomalies to guide the choice. A decade of work has proposed label-free selection criteria for exactly this, from entropy and mass-volume curves [1] to internal relative-evaluation scores [2], meta-learning [3], and rank aggregation across a model pool [4]. Detectors and the selectors that choose

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among them are judged on the same benchmarks, so both inherit whatever those benchmarks get wrong.

Existing benchmarks are contaminated in ways that distort both kinds of comparison. Many anomalies are separable by a single feature or a one-line threshold rule, so every detector scores well and the choice between them barely matters. Some anomalies are exact copies of normal points, labeled both ways because the feature vector does not capture what makes them anomalous, and no detector can ever separate them. Normals leak between train and test when a dataset's duplicate rows are split at random, so a detector is scored on points it has already seen. Some tasks are effectively unsolvable, with no detector beating a random ranking, so a comparison on them carries no signal. Base rates drift so high that the scoring metric divides by a vanishing denominator. And differences are reported as point gaps with no test of significance. A benchmark built on such data measures artifacts, of the detectors and of the methods that choose among them.

We rebuild the benchmark and audit it until it holds. ADReal is a collection of tabular and multivariate time-series datasets produced by one detector-free pipeline whose construction rules each close one contamination channel. On the resulting 151 datasets we evaluate roughly thirty classical and deep detectors under a single consistent preprocessing, scored by a prevalence-aware metric, so no detector is handicapped by an inconsistency in the harness and the comparison is fair across datasets of different anomaly rates. ADReal then serves a second purpose: because a detector must in practice be chosen from normal data alone, the benchmark is also a controlled testbed for unsupervised model selection, where a method picks one detector without labels and is scored by its regret against the oracle-best detector, its criterion computed on held-out normals only. We evaluate eight popular selection methods under this protocol, against two references: a random pick and, more demanding, always deploying the single best-on-average detector.

The leakage-free protocol changes the result. Agreement-based selectors, which dominate under transductive evaluation, fall below a random pick once validation sees only normals. More telling, we measure every selector against a strong reference the literature omits: always deploying the single detector with the best average performance. That fixed detector is hard to beat: even the surviving internal criteria, entropy and mass-volume, are significantly worse than it, and a recent injection-based selector [35] only matches it. We introduce SPARC, which grades detectors on synthetic anomalies built from the normal data, and it is the only label-free method that significantly beats the fixed-detector reference.

Our contributions:

1. **ADReal, a dataset and a protocol for evaluating anomaly detectors.** A construction pipeline that consults no detector removes trivial and mislabeled tasks, closes train/test leakage, caps the base rate, and deduplicates anomalies; two floors computed from the detector pool then drop unsolvable and selection-trivial tasks, so measured performance reflects skill rather than artifacts. From 173 constructed tasks, 151 survive.
2. **An evaluation of a rich set of modern detectors.** We fit roughly thirty classical and deep anomaly detectors under one consistent preprocessing, which gives each dataset its oracle-best detector.
3. **A protocol for evaluating unsupervised model selection, and the finding that the standard criteria fail.** Practice needs the best detector for a specific dataset given only its normal data, a different question from general performance. We score a selection method by its regret against the oracle-best detector, criterion computed on held-out normals only, and against a strong reference the literature omits: always deploying the single best-on-average detector. Under this protocol the agreement selectors (consensus, model-centrality, HITS) fall below a random pick, and even the surviving internal criteria, entropy and mass-volume, are significantly worse than the fixed-detector reference. On ADReal the popular label-free selection methods do not beat simply using one good detector.
4. **SPARC, the one method that beats the reference.** SPARC builds synthetic anomalies from the normal data by resampling features from their own marginals, and grades detectors on how well they separate them. It is the only label-free selector that significantly beats the best fixed detector (regret 0.113 versus 0.130; $p = 0.04$) while beating the internal criteria decisively ($p < 0.001$), with a single probe and no tuning.

2. Related work

Anomaly detectors and benchmark suites. Established detectors span the standard families: isolation-based (Isolation Forest [8]), density- and neighbor-based (LOF [9], KNN), boundary-based (one-class SVM [10]), distribution-based (COPOD [11], ECOD [12], HBOS), projection-based (LODA [13], PCA), and deep one-class and reconstruction models (DeepSVDD [14] and successors [15]). Surveys [15] and large empirical studies agree that no detector dominates across datasets, which is the fact that makes per-dataset selection necessary.

Benchmark suites standardize the comparison: PyOD [6] provides reference implementations; ADBench [5] consolidates 57 tabular datasets and 30 algorithms and reports that no method is statistically best; the DAMI study [16] curates the classic ODDS-derived tabular collection; MacrOData [20] scales tabular outlier detection to thousands of datasets, including the OddBench (real-world semantic-anomaly tables) and OvrBench (one-vs-rest conversions of classification data) collections we draw on; and TSB-AD [19] curates over a thousand time-series under a reliable protocol. These resources standardize detectors and data, but not the protocol under which a label-free selection method is judged.

How detectors are scored, and why the metric matters. Tabular benchmarks report ROC-AUC, average precision (area under precision-recall), and precision-at-n; PyOD and DAMI [6, 16] lead with ROC-AUC and precision-at-n, ADBench [5] with ROC-AUC and AUC-PR. Under the heavy class imbalance of anomaly detection, precision-recall analysis is more informative than ROC [26, 27], but average precision is not comparable across datasets of different anomaly rate: the precision-recall curve moves with prevalence [27], so the same detector scores differently as the base rate changes, and a raw AP gain can be a prevalence artifact rather than a real improvement [29]. Precision-recall-gain [28] rescales precision and recall against the always-positive baseline set by the base rate, which is the principle our metric follows. Time-series evaluation is more fragile still: the widely used point-adjustment protocol inflates scores so badly that a random score can look state of the art [31], and range-based and volume-under-the-surface measures [32, 33] were introduced to restore meaning. We take the lesson that a benchmark must fix one prevalence-aware metric rather than inherit these distortions.

Curation, preprocessing, and filtering in prior benchmarks. Data quality has been addressed before, piecemeal. The DAMI study [16] ships each dataset in normalized and unnormalized, with-duplicate and without-duplicate variants and in several downsampled contamination bands, and shows that normalization, duplicate removal, and outlier ratio each change method rankings, so one raw dataset yields many non-equivalent benchmark instances. Synthetic-anomaly benchmarks depend heavily on how anomalies are injected and only approximate genuine ones under strong assumptions [7]. TSB-AD [19] curates and filters time-series to drop unreliable cases. These works establish deduplication, contamination control, and filtering as necessary, but apply them locally and for detector comparison. ADReal turns them into one detector-free pipeline whose rules each close a specific contamination channel, applied uniformly across sources and aimed at both detector and selection-method evaluation.

Criticism of anomaly-detection benchmarks. A growing literature argues the benchmarks themselves manufacture progress. Wu and Keogh [30] show that popular time-series benchmarks are riddled with triviality, mislabeling, unrealistic anomaly density, and run-to-failure bias, so that a one-line moving average matches deep models. Röchner et al. [34] find that a trivial per-feature extreme-value rule is competitive with state-of-the-art deep detectors on standard tabular benchmarks, and argue evaluation must be rethought. These critiques target detector rankings; the same contamination is even more corrosive for selection, where a trivially separable or mislabeled task makes the choice among detectors meaningless. ADReal is built to remove exactly these channels.

Unsupervised model selection. Choosing a detector or its hyperparameters without labels is the harder problem, because both evaluation and comparison are infeasible without ground truth [3]. Internal criteria score a detector from its own output: excess-mass and mass-volume curves [1] and the internal relative-evaluation score IREOS [2]. Meta-learning transfers performance from a labeled corpus of datasets: MetaOD [3] regresses dataset meta-features to expected performance, ELECT [21] adapts by dataset similarity under a budget, and HYPER [22] uses a hypernetwork to search deep-detector hyperparameters. Closest to our approach, a recent line grades detectors on synthetic anomalies generated from normal data: SWSA [24] builds a pseudo-validation set of generated image anomalies to rank detectors and prompts, and DSV [25] selects self-supervised augmentations by how well they align with the anomaly-generating mechanism. SPARC (Section 6) is in this family, for general tabular and time-series data.

Evaluating selection methods, and consensus. How selection methods are themselves evaluated is contested. A large-scale study of internal strategies [17, 23] concludes that no internal metric reliably beats a naive baseline, selecting models only comparable to a randomly configured detector. A common signal is agreement: averaging or combining detector scores yields a consensus that favors detectors agreeing with the majority [4, 18], and model-centrality and HITS-style rank aggregation formalize this within outlier ensembles [18]. Most of these methods are evaluated *transductively*, computing their criterion on the same data, anomalies included, that supplies the ground truth, which lets a selector's criterion see the very outliers it is asked to find. ADReal computes every selection criterion on held-out normals only and scores each method by regret against the oracle-best detector; the collapse of the agreement selectors we report in Section 5 is a direct consequence.

3. The ADReal benchmark

ADReal is produced by a construction pipeline whose filtering and splitting decisions are made from the data alone, with no detector consulted (Figure 1), so no design choice can be tuned to flatter a detector or a selection method. The pipeline curates raw source datasets into evaluation tasks and applies, in sequence, the contamination controls of Sections 3.2 to 3.5. Three floors are then applied after scoring: two computed from the detector pool (the separability floor and the selection-trivial floor) and one from the data (a minimum of 100 test anomalies). From 174 curated candidate tasks, 173 pass the construction gates and 151 remain after the floors remove 22.

Task structure. Each task is one dataset split three ways: a train set of normals, a validation set of normals, and a test set of held-out normals mixed with hard anomalies. This split carries the benchmark's dual purpose. A detector is fit on the train normals; the test set, the only part that contains anomalies, scores detection quality; and the validation set contains normals only, so a label-free selection method calibrates its criterion exactly as it would in deployment, on normal data alone, without ever seeing an anomaly. No point appears in more than one split (Section 3.3).

3.1. Data sources

ADReal draws from two modalities. Tabular tasks come from ADBench [5], the DAMI collection [16], and the OddBench (real-world semantic-anomaly tables) and OvrBench (one-vs-rest conversions of classification data) collections of MacrOData [20]; multivariate time-series tasks come from TSB-AD [19], windowed into fixed-length feature vectors. A curation stage selects usable source datasets, windows the series, and applies single-feature and permutation triviality pre-filters and a window-purity filter, leaving 174 candidate tasks that enter the pipeline of Figure 1. Table 1 traces the provenance: from the datasets available in each source, how many became candidate tasks and how many survive all contamination controls into the final 151.

Table 1. Provenance of the ADReal benchmark: datasets available in each source, the number selected as candidate tasks by curation, and the number surviving into the benchmark.

Source	Modality	Available	Candidate	In bench.
OvrBench	tabular	754	76	66
OddBench	tabular	187	46	39
TSB-AD	time series	200	41	38
ADBench	tabular	35	7	5
DAMI	tabular	16	4	3
Total		**1,192**	**174**	**151**

3.2. No trivially separable and no unsolvable anomalies

A deployed detector can be staged: a cheap first pass flags the anomalies a simple rule catches, and only the points it clears are passed to a more expensive detector, so the trivial anomalies never reach the second stage. That second stage, which must separate the anomalies the first pass missed, is the one a benchmark should measure and the one a selection method should be chosen for. An anomaly that a single feature or a one-line rule already isolates makes every detector look good and every selector's choice irrelevant, and rewards a detector for work the cheap first stage already does; a benchmark that leaves such anomalies in measures mostly that easy stage. ADReal removes them so that what remains measures the harder stage, where detector choice actually matters. We remove such anomalies with a calibrated OR rule that consults no detector: each feature is thresholded by its own empirical-tail and interior-gap-histogram rarity at a 5% false-positive rate, and a point is flagged if any feature fires, with the same rule repeated in PCA-whitened space to catch oblique separations. For time series, a naive per-channel argmax fires on 56% of random label placements through multiple comparisons, so it is replaced by a permutation test against a circular-shift null at a 6.7% false-positive rate. At the opposite end, a task on which no detector beats a random ranking makes selection meaningless; a **separability floor** removes the 17 tasks whose oracle-best detector reaches base-rate-normalized average precision at or below 0.05, and a **selection-trivial floor** flags tasks on which every detector performs within a hair of the best, where any pick is a good pick; the 2 tasks it flags also fail the separability floor, so these two floors together with the minimum-data floor of Section 3.4 remove 22 tasks in all. Figure 2 illustrates both ends of this filtering: datasets whose anomalies are all trivially separable are dropped, and within the datasets that survive, the easy anomalies are removed while the anomalies that remain are interspersed with the normals. Figure 3 shows the same separation in the filter's severity statistic.

3.3. No mislabeled anomalies and no leakage

Three failures let a selector be scored on the wrong answer. First, windowing mislabels: a sliding window inherits an anomaly flag from a single touched point, turning long stretches of normal signal into false anomalies. A stage-zero filter drops any series where fewer than half of anomaly points fall in contiguous segments of at least half a window. Second, label ambiguity: in coarse tabular data the same feature vector is sometimes labeled normal and sometimes anomalous, because

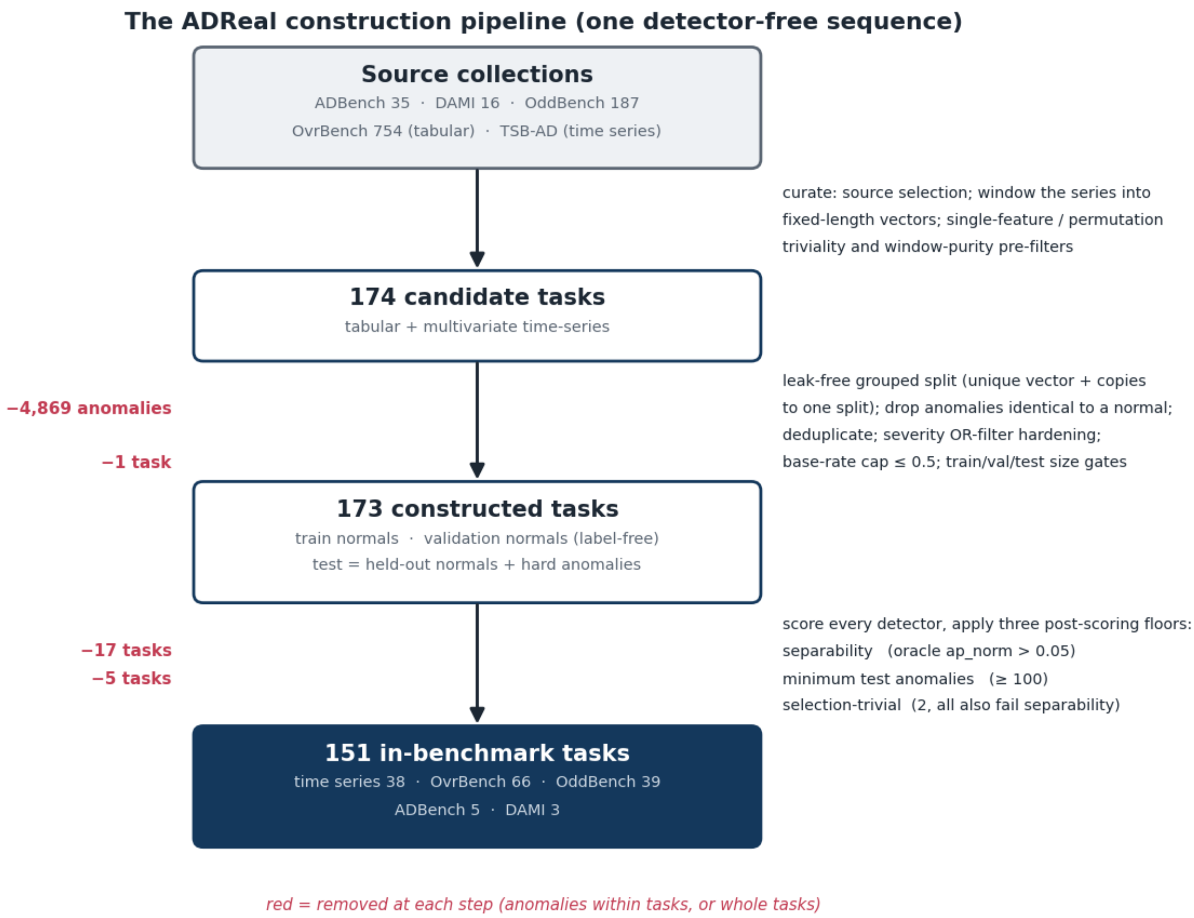

Figure 1. The ADReal construction pipeline. One detector-free sequence curates source collections into 174 candidate tasks, applies per-task contamination controls to reach 173 constructed tasks, and removes 22 tasks with three post-scoring floors, leaving 151. Numbers in red are the tasks or anomalies removed at each step.

the features do not capture what makes it anomalous; such an anomaly is an exact copy of a normal point and is unlabelable, so we remove every anomaly identical to a normal. Across the candidate tasks this removes 4,869 such anomalies in 32 tasks. Third, leakage: a naive random split of the normals leaks, because outlier collections contain many duplicate rows; on the source data it puts a normal into two splits in 62 of 173 tasks, 8,285 points in all, so a detector would be scored on rows it trained on. We instead split the normals 60/20/20 into train, validation, and test by a **grouped split** that assigns each unique feature-vector, together with all its duplicate copies, entirely to one split, so no point ever appears in two splits while the density of repeated values is preserved within each split; for time series a contiguous-block split plus purging of boundary-straddling windows serves the same end. Every label-free selection criterion is computed on the validation split, which contains normals only, so no selector observes a test anomaly.

Throughout, a duplicate is an exact match at float32 precision, each row cast to float32 and rounded to six decimals, not a similarity threshold: duplicate normals are kept but grouped to a single split, anomalies identical to a normal are removed, and repeated anomalies are collapsed to distinct cases (Section 3.5). This contamination is widespread in the source data. Across the datasets these benchmarks provide, 49% contain duplicate normal rows, 35% duplicate anomalies, and 24% contain anomalies that are exact copies of a normal. Table 2 shows the range on representative datasets, from clean to severe: `donors` repeats one normal row 93,960 times, and `http` has 2,211 anomalies of only 72 distinct values, 2,092 of them exact copies of a normal.

Table 2. Duplication in representative source datasets: normal rows and their distinct values (with the largest group of identical normals), anomaly rows and their distinct values, and the number of anomalies that are exact copies of a normal (float32-precision matching).

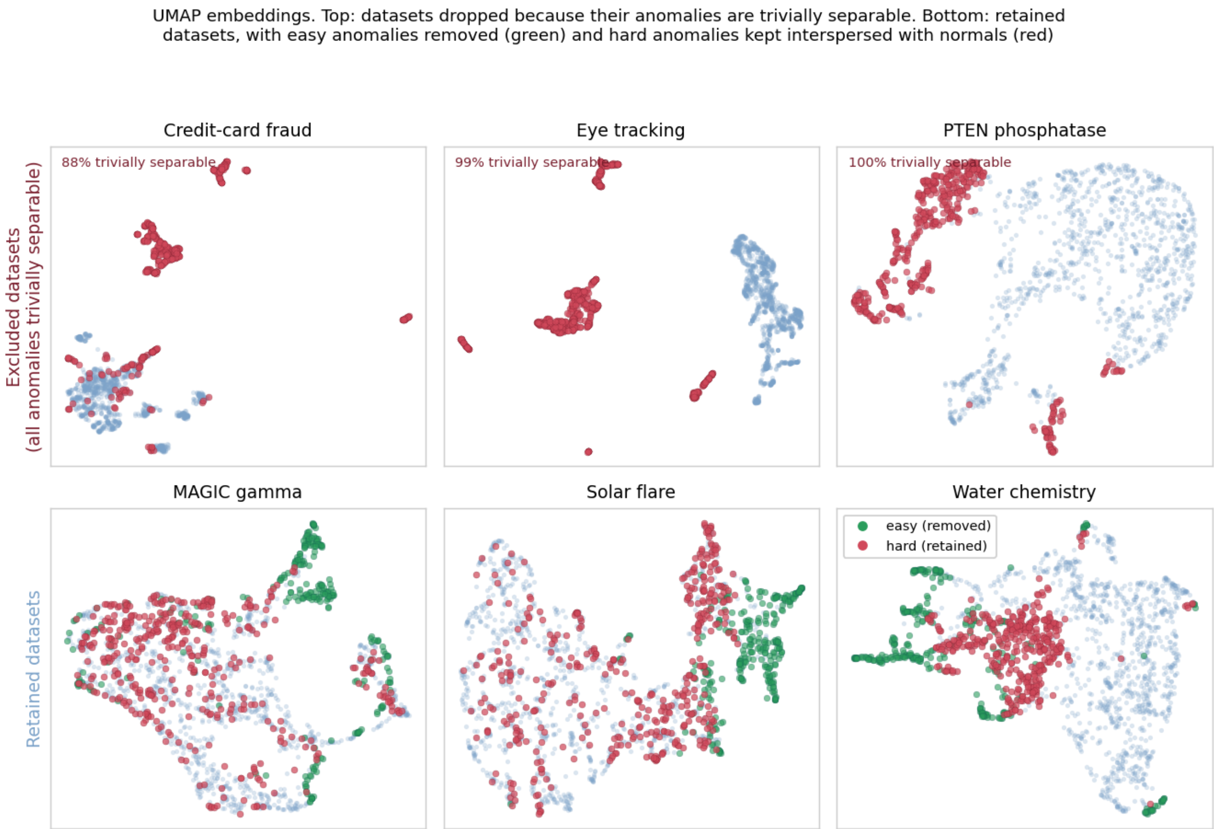

Figure 2. The triviality controls, shown as UMAP embeddings of the raw source data. Top: three datasets dropped from the benchmark because nearly all their anomalies (red) are trivially separable from the normals. Bottom: three retained datasets, where the removed easy anomalies (green) form their own separated clusters while the retained hard anomalies (red) are interspersed with the normals.

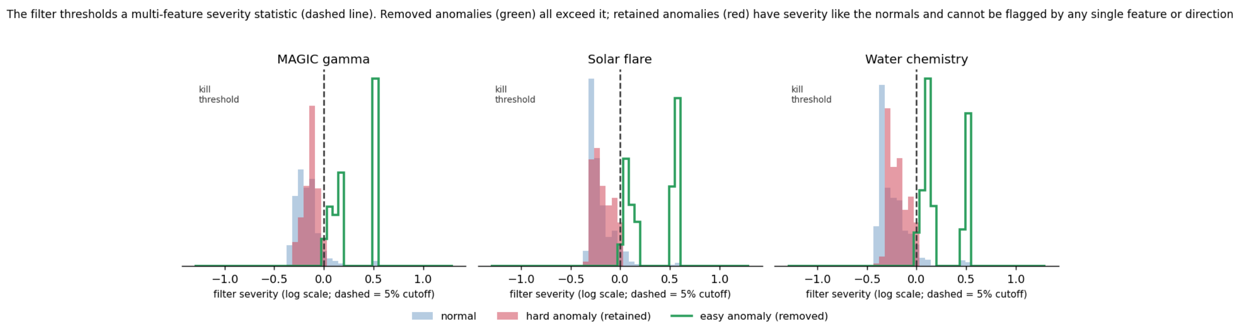

Figure 3. The same effect in one dimension: the hardening filter thresholds a multi-feature severity statistic (dashed line, the 5% cutoff). The removed easy anomalies (green) all exceed it, while the retained hard anomalies (red) have severity like the normals and so cannot be flagged by any single feature or direction. Because the statistic aggregates all features and whitened directions, an anomaly separable on any one of them is removed even when it overlaps the normals along another.

3.4. One consistent KPI with a bounded base rate

Every detector is scored by a single quantity, base-rate-normalized average precision, ap_norm = $(\mathrm{AP} - r)/(1 - r)$ where r is the task's anomaly rate, comparable across tasks of different base rates. Because ap_norm divides by $1 - r$, a task whose test set is mostly anomalies makes the metric degenerate; a **base-rate cap** holds the test anomaly fraction at or below one half by subsampling hard anomalies to parity with the held-out normals, leaving a median test anomaly fraction of 0.35 across the benchmark. A task must also retain enough anomalies to estimate the metric, so a floor of at least 100 test anomalies removes 5 further tasks. A selection method is then evaluated by its regret, the ap_norm of the oracle-best detector minus the

Dataset (source)	Normals	Distinct	Largest group	Anomalies	Distinct	= a normal
donors (ADBench)	582,616	27,754	93,960	36,710	7,667	0
http (ADBench)	565,287	221,830	295	2,211	72	2,092
poker-hand (OvrBench)	513,701	147,838	16	111,845	86,985	410
ProtestViolence (OddBench)	17,538	15,433	13	2,686	2,423	19
Cardiotocography (ADBench)	1,655	1,647	4	176	175	0
Annthyroid (DAMI)	6,595	6,595	1	534	534	0

ap_norm of the detector it picked.

3.5. *No duplicated anomalies and one significance test*

Repeated near-identical anomalies inflate whichever detector happens to catch the template. Datasets are deduplicated across sources by name and dimensionality, and within a task the anomaly set is reduced to distinct cases. Selection methods are compared by a paired Wilcoxon signed-rank test on per-task regret, reported two-sided, together with the median and tail of the regret distribution. A raw difference of means that a handful of easy tasks could produce is never reported as a result.

3.6. *What the benchmark looks like*

The 151 tasks are not uniform, and the two modalities differ in ways that matter for both detection and selection (Figure 4, Table 3). Tabular tasks range from 2 to over 300 features and sit near the base-rate cap, with a median test anomaly fraction of 0.43; their oracle detector is comparatively strong (median ap_norm 0.30). Time-series window features are higher-dimensional (median 16, up to 408), have a much lower prevalence (median 0.15), and are harder: the oracle detector reaches a median ap_norm of only 0.19. This spread is deliberate. The contamination controls remove the tasks on which selection would be meaningless, but leave a benchmark that is genuinely difficult, with an oracle far from perfect on most tasks, which is the regime in which a selection method's choice actually matters.

Table 3. The 151 ADReal tasks by source: modality, number of tasks, and the median dimensionality, test anomaly fraction, and oracle-best detector ap_norm.

Source	Tasks	Dim (med.)	Anomaly frac. (med.)	Oracle ap_ norm (med.)
OvrBench (tabular)	66	10	0.48	0.29
OddBench (tabular)	39	7	0.35	0.28
TSB-AD (time series)	38	16	0.15	0.19
ADBench (tabular)	5	36	0.50	0.51
DAMI (tabular)	3	21	0.23	0.65

4. Detectors and their evaluation

Detector pool. A selection method chooses from a fixed pool of 32 detectors that spans the classical families and a set of deep one-class and reconstruction models (Table 4). The classical part is the standard PyOD [6] set: Isolation Forest [8], LOF [9], KNN, ECOD [12], COPOD [11], HBOS, PCA, CBLOF, and LODA [13]. For the four algorithms whose single most influential hyperparameter drives their behavior, the pool sweeps a small grid of that hyperparameter rather than one default, so selection includes hyperparameter choice and not only algorithm choice: neighborhood size for LOF and KNN, histogram bins for HBOS, and retained variance for PCA. The deep part adds one detector each of DeepSVDD [14], an autoencoder, a variational autoencoder, internal contrastive learning [37], deep isolation, robust collaborative autoencoding, and scale learning [38]. Every detector, classical and deep, is fit on the train normals and scored on the same standardized features, so no detector is handicapped by a preprocessing inconsistency; the oracle-best detector per task defines the zero of regret.

Table 4. The 32-detector pool. For the four classical algorithms most sensitive to one hyperparameter, the pool sweeps a grid of that hyperparameter, so a selection method chooses configuration as well as algorithm.

Detector	Family	Hyperparameter swept	Variants
LOF [9]	Density / neighbor	neighborhood size k	6
KNN	Density / neighbor	neighborhood size k	6
HBOS	Distribution	histogram bins	4
PCA	Projection	retained variance	4
Isolation Forest [8], CBLOF	Isolation, density	default	2
ECOD [12], COPOD [11]	Distribution	default	2
LODA [13]	Projection	default (seeded)	1
DeepSVDD [14] and six more	Deep	one config each	7

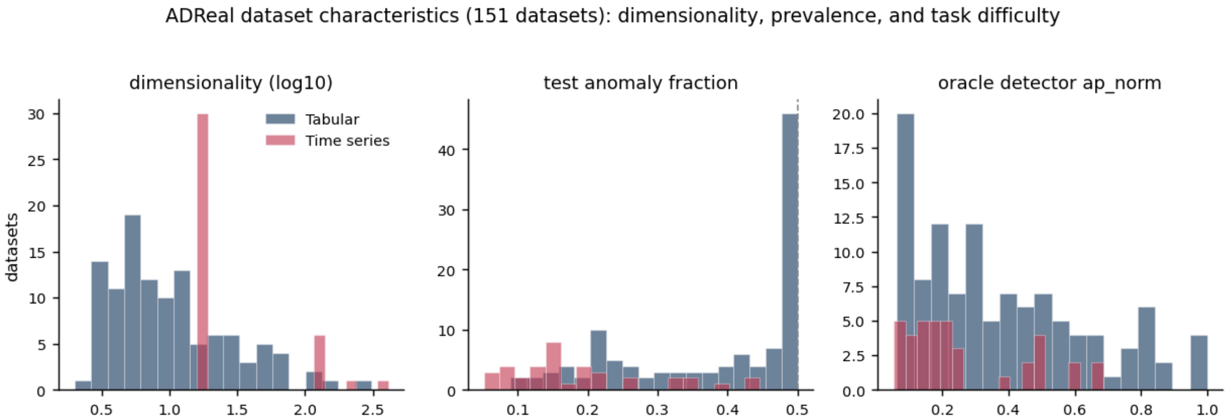

Figure 4. Characteristics of the 151 ADReal datasets, split by modality: dimensionality (log scale), test anomaly fraction (dashed line marks the base-rate cap at 0.5), and the difficulty of each task measured by its oracle-best detector. Time-series tasks are higher-dimensional, lower-prevalence, and harder than the tabular tasks.

No single detector dominates, and deep does not beat classical. The detector evaluation is the substrate for every selection result: on each task it fixes the oracle-best detector that defines the zero of regret, and no detector holds that role everywhere. Figure 5 ranks all thirty-two detectors by mean base-rate-normalized average precision across the 151 datasets. Neighbor and density methods (LOF and KNN variants) take the top six places; the best deep detector, DeepSVDD, is individually strong but ranks seventh by mean. Classical detectors supply the per-dataset oracle on 118 of 151 datasets (78%) against 33 for the deep pool, and the best deep detector per task trails the best classical by 0.071 on average, so adding the entire deep pool to the classical one raises the oracle by only 0.021. Fairly evaluated under one consistent preprocessing, classical detectors carry this benchmark; the deep pool shifts each dataset's oracle by only 0.021 on average, so the selection results below are essentially unchanged with or without it.

Grouping detectors by family (Table 5) shows why no family is dispensable and why a family's average is a poor guide to when it wins. The neighbor and density family supplies almost half of all per-dataset oracles. The distribution family has a low average yet supplies the oracle on nearly a quarter of datasets, because a few of its members win outright on specific datasets; the isolation family has a respectable average but is never the single best.

Table 5. Detector families on the 151-dataset ADReal benchmark: number of detectors in the pool, the share of datasets on which the family supplies the oracle-best detector, and the family's mean base-rate-normalized average precision (ap_norm).

Family	Detectors	Oracle-best share	Mean ap_norm
Density / neighbor	13	45%	0.166
Distribution	6	24%	0.087
Deep	7	22%	0.123
Projection	5	9%	0.072
Isolation	1	0%	0.106

5. Unsupervised model selection: methods, protocol, and results

Selection methods and references. We evaluate eight label-free selectors, all restricted to the leakage-free protocol. *Entropy* (excess-mass) and *mass-volume* [1] score each detector by the shape of its score distribution over the data and a uniform reference sample. *Consensus* [4], *model-centrality*, and *HITS* rank detectors by agreement with the pool [18]. *UDR* aggregates stability across configurations, *IFOREST-R* is a random-Isolation-Forest reference [17], and *IREOS* [2] scores the separability of the top-flagged points. We add the injection-based selector of Goswami et al. [35], which grades detectors on synthetic anomalies, as an external competitor, and SPARC (Section 6). Two references bound the field. A uniform-random pick over the pool is the floor: a selector that cannot beat it carries no information. The bar is the *best fixed detector*, always deploying the single detector with the lowest average regret across the benchmark (LOF with $k = 10$), which uses no per-task information at all. A selection method earns its keep only by beating that fixed detector, a baseline prior work omits.

Protocol. Every selection method receives the same inputs and is scored the same way. A detector is fit on the train normals; the method computes its criterion using only the

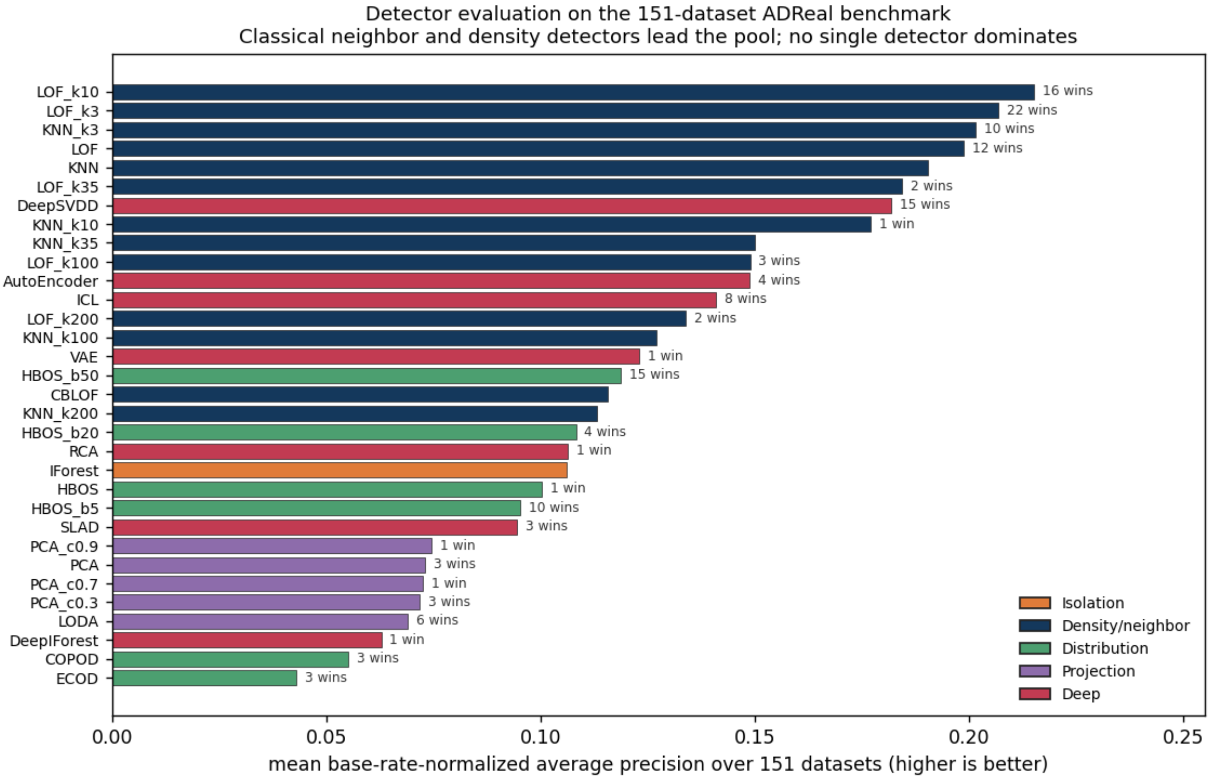

Figure 5. Detector evaluation on the 151-dataset ADReal benchmark: mean base-rate-normalized average precision per detector (higher is better), colored by family, with the number of datasets on which each detector is the oracle-best. Neighbor and density detectors lead; classical detectors supply the oracle on 78% of datasets, and the deep pool adds only 0.021 to the average oracle.

validation normals and the detectors' scores on them, with no access to any anomaly; it returns one chosen detector; and we score that choice by regret, the gap in test ap_norm between the oracle-best detector and the chosen one. Regret is aggregated over tasks and methods are compared by the paired Wilcoxon test of Section 3.5. Because the criterion never sees an anomaly, this measures label-free selection as it is actually deployed, unlike the transductive protocol under which these methods usually report gains.

Adding deep detectors to the candidate pool raises every selector's regret. A selector's candidate pool is itself a design choice, so we evaluate every method on three pools: the 25 classical detectors, the 7 deep detectors, and the full 32 (Table 6). Enlarging the classical pool to include the deep detectors raises regret for every label-free method, SPARC from 0.113 to 0.162, Goswami from 0.122 to 0.151, entropy from 0.163 to 0.176, mass-volume from 0.165 to 0.174, and the agreement selectors similarly; the increases are significant for SPARC, Goswami, entropy, mass-volume, and HITS ($p < 0.01$). Selecting among the deep detectors alone is worse still, 0.169 to 0.259, at or below the random floor. The deep detectors are the cause. They separate each method's synthetic or distributional signal on the validation normals without generalizing to the test anomalies: averaged over tasks a deep detector earns a higher probe grade than a classical one (separation ROC-AUC 0.73 against 0.64) while reaching no higher ap_norm on the real anomalies (0.12 for both), so a selector that may pick deep detectors is drawn to them by an inflated grade and then pays regret. A fair comparison across methods therefore holds the candidate pool fixed, and every method performs best on the classical pool, where the results below are stated.

On the matched classical pool, only SPARC beats the fixed detector. The agreement selectors (consensus, HITS, model-centrality, 0.218 to 0.223) do not beat a random pick (0.219): with no anomalies in validation, the quantity they rank on is dominated by detector behavior on normals and does not track detection skill. Entropy and mass-volume clear the random floor (0.163 and 0.165) but fall short of the fixed detector at 0.130. Goswami's injection selector reaches 0.122, matching the fixed detector but not beating it (p = 0.88). SPARC is the only method that significantly beats the fixed detector, at regret 0.113 (p = 0.04). On the full 32-detector pool no label-free method beats the fixed detector, and SPARC (0.162) is edged by Goswami (0.151): a Borda vote over five injection types is more robust to a single overfitting deep candidate than SPARC's single probe.

Table 6. Selection regret (lower is better) by candidate pool, with the best-fixed-detector and random references. Adding the

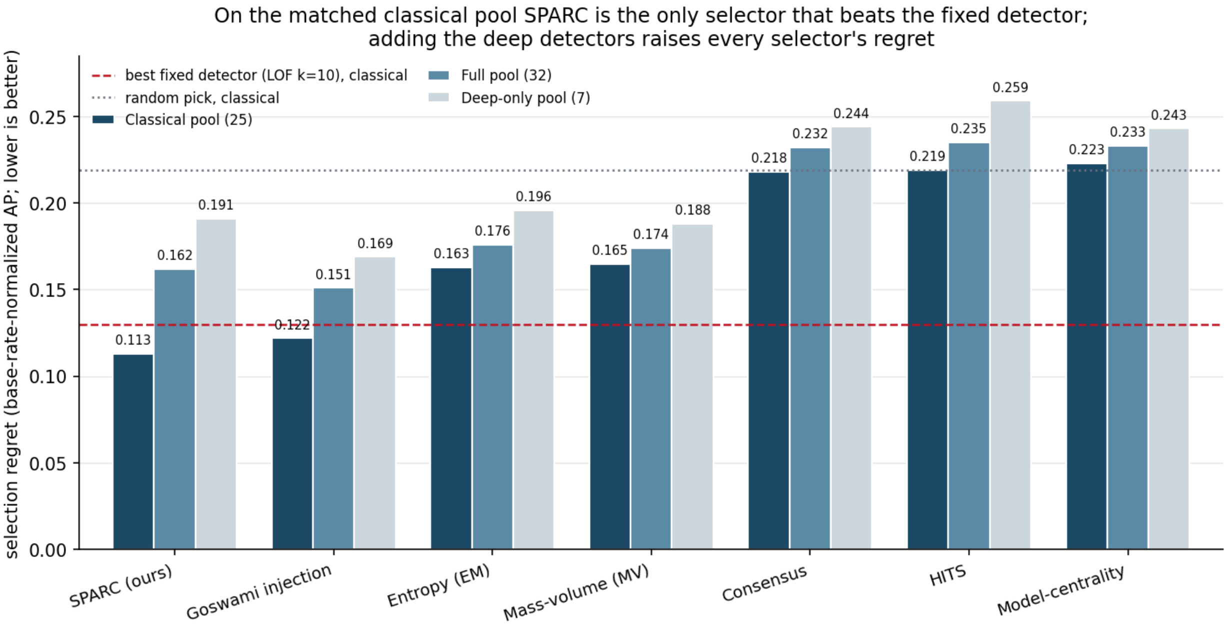

Figure 6. Model-selection regret on the 151-task ADReal benchmark (base-rate-normalized average precision; lower is better), per method on the classical (25), deep-only (7), and full (32) detector pools. The dashed line is the best fixed detector (LOF, $k = 10$); the dotted line is a random pick. On the matched classical pool SPARC is the only method that beats the fixed detector; adding the deep detectors raises every selector's regret.

deep detectors raises every selector's regret; on the matched classical pool SPARC is the only method that beats the fixed detector.

Selector	Classical (25)	Deep (7)	Full (32)
SPARC (Section 6)	**0.113**	0.191	0.162
Goswami injection [35]	0.122	0.169	0.151
Best fixed detector	0.130	0.163	0.130
Entropy (EM)	0.163	0.196	0.176
Mass-volume (MV)	0.165	0.188	0.174
Consensus	0.218	0.244	0.232
HITS	0.219	0.259	0.235
Model-centrality	0.223	0.243	0.233
Random pick	0.219	0.223	0.220

Selection helps only where no detector is near-universal. Breaking the classical-pool regret down by source (Table 6b) locates where selection adds value over the fixed detector. On time series and OvrBench a single neighbor detector is already near the oracle (fixed-detector regret 0.038 and 0.145), and no method meaningfully beats it. Selection earns its keep on OddBench, where the oracle detector varies from dataset to dataset: there SPARC reaches 0.153 against the fixed detector's 0.198 ($p = 0.029$) and entropy's 0.164. The agreement selectors collapse on both tabular collections (0.23 to 0.25) and only sit near random on time series, while entropy and mass-volume trail on time series (0.145 and 0.155), where a neighbor detector is near-oracle.

Table 6b. Classical-pool selection regret by data source (lower is better). SPARC's advantage over the fixed detector is concentrated in the heterogeneous tabular collections; on time series a neighbor detector is already near-optimal.

Selector	OvrBench (66)	OddBench (39)	TSB-AD / TS (38)	All (151)
SPARC (Section 6)	0.137	**0.153**	0.037	**0.113**
Goswami injection [35]	0.137	0.159	0.053	0.122
Best fixed detector	0.145	0.198	0.038	0.130
Entropy (EM)	0.162	0.164	0.145	0.163
Mass-volume (MV)	0.162	0.170	0.155	0.165
Consensus	0.243	0.234	0.126	0.218
Random pick	0.237	0.241	0.141	0.219

6. SPARC: selection by a synthetic probe

Motivation. The surviving label-free criteria score a detector from the shape of its output on normal data alone; none confronts the detector with an anomaly. But a detector's job is to separate anomalies from normals, and different detectors separate different kinds of anomaly. SPARC makes this concrete: it manufactures synthetic anomalies from the normal data and selects the detector that best finds them, never observing a real anomaly, so it runs under the same normals-only protocol as the baselines. Grading detectors on injected synthetic anomalies has been used for time-series selection before [35]; SPARC uses a single feature-corruption probe that applies to tabular and time-series data alike.

The probe. From the validation normals, SPARC builds one kind of synthetic anomaly. For each normal point it resamples a random subset, about 40%, of its features, drawing each from that feature's own observed values, so every coordinate stays a value the data actually takes while the combination violates the normal joint structure [36]. This is the feature-corruption operator of self-supervised tabular learning (SCARF [39], VIME [40]), here repurposed not to train a detector but to grade one: a detector that separates these dependency-violating points from the normals is sensitive to exactly the local, joint structure that real anomalies inhabit. Each detector is graded by the ROC-AUC with which its scores separate the probe from the validation normals, and the highest-graded detector is SPARC's choice.

Candidates. Deep detectors overfit the synthetic probe, separating it without generalizing to real anomalies, so SPARC selects among the classical detectors while the pool and the oracle still include the deep detectors. This is not specific to SPARC: Section 5 shows that adding the deep detectors raises regret for every label-free selector, because they earn an inflated probe or injection grade that their detection performance does not support. That is the whole method: one probe, one grade per detector, an argmax, with a single parameter (the resample fraction) and no thresholds to tune.

What matters. One design choice is not obvious and earns the result (Table 7): drawing the resampled value from the feature's *observed* values rather than a continuous interpolation. On discrete features an interpolated value never occurs in the data, which lets marginal detectors trivially flag the probe and be wrongly selected; the observed-value draw removes that artifact and lowers regret from 0.122 to 0.113. The classical-candidate restriction earns the rest: with deep detectors eligible to win the probe, regret rises as they overfit the synthetic anomalies.

Table 7. SPARC ablation (mean regret over the 151 tasks, lower is better).

SPARC variant	Regret
SPARC (single probe, observed-value draw, classical candidates)	**0.113**
Resampling by bin interpolation instead of observed values	0.122
Best fixed detector (reference)	0.130

SPARC is the only selector that beats the fixed detector. SPARC reaches regret 0.113, beating the best fixed detector (0.130) at $p = 0.04$ and, on the matched classical pool, the internal criteria entropy and mass-volume (0.163, 0.165) at $p < 0.001$ (Figure 6). Goswami's injection selector (0.122) matches the fixed detector but does not beat it. No method built on the shape of the normal-data score distribution clears the fixed-detector bar; the only method that clears it grades detectors on synthetic anomalies. The result does not hinge on the deep detectors: as Section 4 shows, a fairly preprocessed classical pool supplies almost every dataset's oracle.

Selection earns its keep where no detector is near-universal. Figure 7 breaks the SPARC result down by modality against the fixed detector. On time series and OvrBench a single neighbor detector is already near the oracle (0.038, 0.145), and SPARC ties it (0.037, 0.137) while the internal criteria trail badly on time series (entropy 0.155). On OddBench, where the oracle detector varies from dataset to dataset, SPARC's per-task choice pays off: 0.153 against the fixed detector's 0.198 ($p = 0.029$) and entropy's 0.193. Selection helps precisely where a fixed detector cannot, and costs nothing where one suffices.

7. Analysis and discussion

Where selection helps. SPARC's value over a fixed detector follows how much the oracle detector varies across a collection. On time series and OvrBench one neighbor detector is near-oracle almost everywhere, so a fixed choice is already near-optimal and no label-free method, SPARC included, can add much. On OddBench the oracle detector changes from dataset to dataset, and there SPARC's probe ranks detectors well enough to beat the fixed detector. The synthetic probe is a good proxy for real anomalies exactly when those anomalies are local, joint-structure violations, which is what the benchmark's hardening leaves behind.

What the hardening selects for. Removing trivially separable anomalies means removing marginally extreme ones: the dropped anomalies deviate along a single feature or a linear rule, while the retained anomalies are marginally typical, with per-feature values that look normal, yet sit in locally sparse joint regions. So the benchmark's hard anomalies favor neighbor and local-density detectors over purely marginal ones (Section 4), and correspondingly favor a probe that tests joint structure, which also explains why the marginal internal criteria fare poorly here.

What the probe sees and misses. The synthetic probe grades a detector on one kind of anomaly, and its selection bias follows from that (Table 8). Across the 151 tasks the probe ranks a deep detector first on 80 but a deep detector holds the oracle on only 33: a dependency-breaking probe is exactly what a reconstruction or one-class deep model is built to flag, so it separates the probe whatever the model's skill on real anomalies (Section 5), which is why SPARC selects among the classical detectors. In the other direction the probe ranks a distribution detector first on a single task while that family holds the oracle on 36, because resampling each feature from

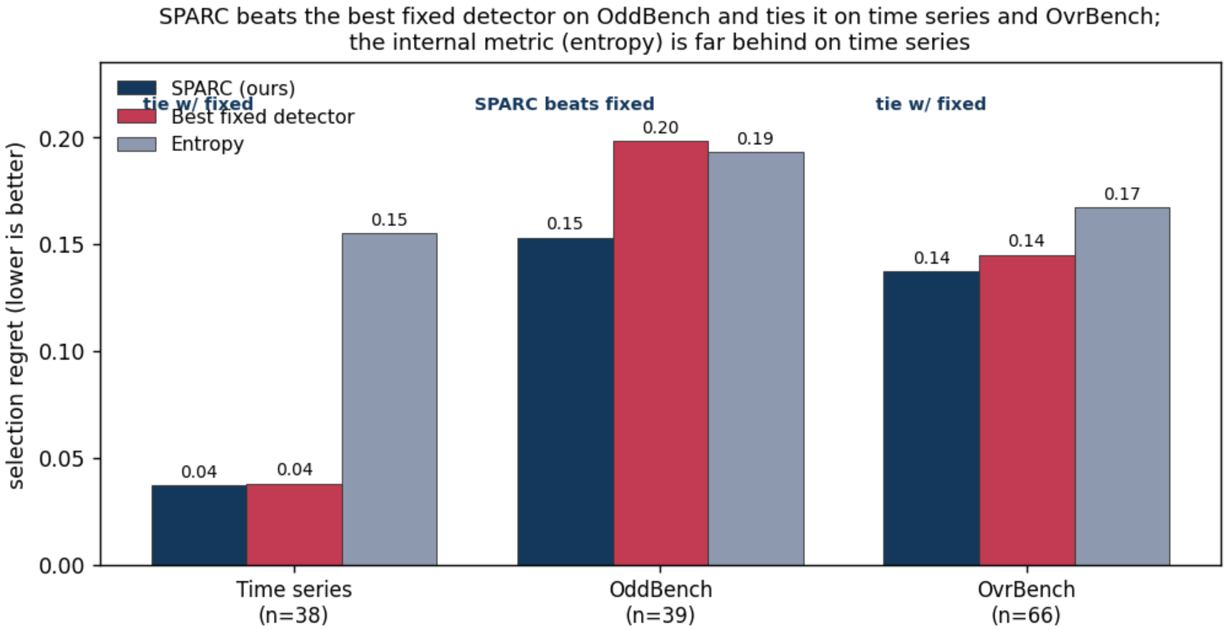

Figure 7. Selection regret by modality for SPARC versus the best fixed detector and entropy. SPARC significantly beats the fixed detector on OddBench ($p = 0.029$) and ties it on time series and OvrBench; the internal criterion is far behind on time series. Significance is the paired Wilcoxon test of SPARC against the fixed detector.

its own observed values leaves the per-feature marginals intact and gives a marginal detector nothing to read. The probe is matched to the neighbor family, which it ranks first on 70 tasks against 68 oracles. SPARC beats the fixed detector where a neighbor detector is the oracle and the probe can see it, and a probe that violates marginals rather than joint structure is the natural complement for the marginal regime (Section 8).

Table 8. Which detector family the synthetic probe ranks first, against which family holds the oracle-best detector, over the 151 tasks with all 32 detectors eligible (the oracle column matches Table 5). The probe over-ranks the deep family, does not see the distribution family, and is matched to the neighbor family.

Detector family	Probe ranks #1	Holds the oracle
Density / neighbor	70	68
Distribution	1	36
Projection	0	14
Isolation	0	0
Deep	80	33

Why the references matter. Two references reshape the picture that internal-metric numbers alone would paint. Against a random pick, entropy and mass-volume look like the methods to beat; against a fixed good detector, they are worse than using no selection at all, and the only method that clears the bar confronts detectors with synthetic anomalies. Constructing the benchmark also surfaced contamination in the source data that quietly moves results, from anomalies identical to normals to duplicate rows leaked across the split, all of which the pipeline removes.

8. Limitations

ADReal covers tabular and multivariate time-series data; univariate series and other modalities are out of scope here, and the construction rules would need modality-specific instantiation to extend. On this data SPARC's win over the best fixed detector is 0.113 against 0.130, and it comes almost entirely from OddBench, while on time series and OvrBench a fixed neighbor detector already suffices. Selection adds value only where the oracle detector varies across a collection. The hardening filter, by removing marginally separable anomalies, tilts the benchmark toward local-density anomalies and thus toward neighbor detectors and joint-structure probes; a benchmark built to retain marginal anomalies would test a different regime, and a probe targeting marginal gaps is the natural complement to SPARC there. The pool of detectors is finite, so the oracle that defines regret is the best *available* detector, not the best possible one.

9. Conclusion

Benchmarks for anomaly detection, and for the label-free methods that select among detectors, have been measuring artifacts: trivial and unsolvable tasks, anomalies that are copies of normals, leaked normals, degenerate base rates, and untested differences. ADReal rebuilds the measurement with a detector-free pipeline, audited until each channel is closed, then evaluates detectors under one prevalence-aware protocol and selection methods by regret against the oracle-best detector and against a strong fixed-detector reference. The reference reframes the field: the popular internal selection criteria are worse than simply deploying one good detector, and the agreement selectors fall below random. SPARC, which grades detectors on a single synthetic probe of the normal data, is the only label-free method that significantly beats the fixed detector, and it does so where it matters, on the collection whose oracle detector varies, with one probe and no tuning.

Data and code availability

The benchmark construction pipeline, the per-task detector scores, the selection-method implementations, and the figures are released at the project repository. A permanent archive with a citable DOI accompanies the camera-ready version.

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